Received: 13 March 2020 | Accepted: 29 May 2020

DOI: 10.1002/jcp.29874

ORIGINAL RESEARCH ARTICLE

Journal of Cellular Physiology WILEY

Gene interaction network approach to elucidate the multidrug resistance mechanisms in the pathogenic bacterial strain *Proteus mirabilis*

Sravan K. Miryala | **Anand Anbarasu** | **Sudha Ramaiah**

Medical and Biological Computing Laboratory, Department of Bio-Sciences, School of Biosciences and Technology, Vellore Institute of Technology, Vellore, India

Correspondence
Sudha Ramaiah, Medical and Biological Computing Laboratory, Department of Bio-Sciences, School of Biosciences and Technology, Vellore Institute of Technology, Vellore 632 014, India.
Email: sudhaanand@vit.ac.in

Funding information
Indian Council of Medical Research, Grant/Award Number: IRIS ID: 2019-0810

Abstract

Proteus mirabilis is one among the most frequently identified pathogen in patients with the urinary tract infection. The multidrug resistance exhibited by *P. mirabilis* renders the treatment ineffective, and new progressive strategies are needed to overcome the antibiotic resistance (AR). We have analyzed the evolutionary relationship of 29 *P. mirabilis* strains available in the National Center for Biotechnology Information-Genome database. The antimicrobial resistance genes of *P. mirabilis* along with the enriched pathways and the Gene Ontology terms are analyzed using gene networks to understand the molecular basis of AR. The genes *rpoB*, *tufB*, *rpsl*, *fusA*, and *rpoA* could be exploited as potential drug targets as they are involved in regulating the vital functions within the bacterium. The drug targets reported in the present study will aid researchers in developing new strategies to combat multidrug-resistant *P. mirabilis*.

KEYWORDS

clustering analysis, functional enrichment analysis, gene ontology, *Proteus mirabilis*, two-component system

1 | INTRODUCTION

Indiscriminate use of antibiotics over several decades has culminated in the development of multidrug resistance. The problem of antibiotic resistance (AR) alarms the global health regulatory bodies and researchers are working tirelessly to address the concern. The World Health Organization (WHO) has listed out a wide range of organisms that require a varying degree of attention in terms of AR research (Nathan & Cars, 2014). *Proteus mirabilis* is reported as a high priority organism by WHO; it is a Gram-negative enteric bacteria that cause urinary tract infections (UTIs) in humans, predominantly in individuals with Type 2 diabetes and indwelling urinary catheters (Schaffer & Pearson, 2015).

P. mirabilis has been isolated frequently from clinical samples of patients with UTI. Individuals with impaired urinary tracts due to spinal cord injury or anatomic abnormality are more prone to the frequent occurrence of *P. mirabilis* associated UTI. *P. mirabilis* strains with its strong adherence ability, can infect the kidney, bladder and can even penetrate the bloodstream; thus it can also cause bacteremia, fever, acute pyelonephritis, and sepsis (Hammad, Sulaiman, Aziz, & Noor, 2019; Mihankhah, Khoshbakht, Raeisi, & Raeisi, 2017; Pearson et al., 2008; Schaffer & Pearson, 2015; Sharma, Preston, & Hage, 2019). Multidrug resistance exhibited by the *P. mirabilis* renders the treatment unproductive resulting in severe disease conditions (Guo et al., 2019; Hammad, Sulaiman, Aziz, Noor, et al., 2019).

For the present study, we have collected the whole genome sequences of *P. mirabilis* strains from the National Center for Biotechnology Information (NCBI)-Genome database and constructed the phylogenetic tree to understand the evolutionary relationship between the different strains of *P. mirabilis*. We have also collected the resistance genes in *P. mirabilis* strains along with their AR profiles. *P. mirabilis* HI4320 is considered as a low-passage clinical isolate and regarded as the representative strain of the *P. mirabilis* (Mobley & Chippendale, 1990). The antimicrobial resistance (AMR) genes in *P. mirabilis* HI4320 strain were curated and a gene interaction network was constructed to understand the multidrug resistance patterns. Gene interaction network studies have recently gained the attention of researchers and are considered to be useful approaches to understand multidrug resistance in pathogenic and opportunistic bacterial spp, and other biological abnormalities (Anitha, Anbarasu, & Ramaiah, 2014, 2016; Anitha, Bag, Anbarasu, & Ramaiah, 2015; Bag, Ramaiah, & Anbarasu, 2015a, 2015b; Debroy, Miryala, Naha, Anbarasu, & Ramaiah, 2020; Miryala, Anbarasu, & Ramaiah, 2019, 2020a; 2020b; Miryala & Ramaiah, 2019; Naha, Kumar Miryala, Debroy, Ramaiah, & Anbarasu, 2020; Parimelzaghan, Anbarasu, & Ramaiah, 2016). The interaction data for the targeted AMR genes extracted from the dedicated interaction databases (Miryala, Anbarasu, & Ramaiah, 2018). Gene interaction network of AMR genes was constructed and various topological parameters and the functional relatedness of AMR genes with the other functional partners were analyzed using clustering analysis. Our results help in better understanding of the molecular basis of multidrug resistance mechanisms in *P. mirabilis*. The genes listed as potential drug targets can be exploited for developing new compounds that can be used for therapeutic applications to control *P. mirabilis* infections.

2 | TOOLS AND DATABASES

2.1 | NCBI genome

NCBI genome database (https://www.ncbi.nlm.nih.gov/genome) consists of whole-genome sequences of more than 1,000 organisms, which includes the whole genome sequences of the organisms and also organisms for whose sequencing is in progress. The database consists of the sequences related to three main domains of life, such as bacteria, archaea, and eukaryote, along with the sequences related to many viruses, bacteriophages, viroids, plasmids, and organelles.

2.2 | ParSNP

ParSNP (https://github.com/marbl/parsnp) is a rapid genome multi-alignment tool designed to align the genome sequences. User can give draft assemblies or the finished genomes as an input. The tool aligns and provides the output as the multiple sequence alignment of given sequences, single nuclear polymorphism (SNP) variations, and the core genome phylogeny. Visualization tools can be used to visualize and analyze the phylogenetic tree (Treangen, Ondov, Koren, & Phillippy, 2014a, 2014b).

2.3 | Interactive Tree Of Life (iTOL)

iTOL (https://itol.embl.de/) is an online tool used for visualization, annotation, and management of phylogenetic tree analysis. It facilitates the user with the options to explore the trees directly in the web browser and annotate it. The user can adjust, label, change style or fonts. It also provides the options to export the outputs in high-quality vector or the graphical bitmap formats (Letunic & Bork, 2019).

2.4 | ResFinder

ResFinder (https://cge.cbs.dtu.dk/services/ResFinder/) is an online tool that helps in capturing the AMR genes from the whole genome sequence. It contains two python scripts that are used to identify acquired AMR genes (ResFinder.py) and for finding the chromosomal mutation (PointFidner.py). The database uses inputs as a full sequence, partial sequence, or short sequence reads from various sequencing platforms and searches the AMR genes using Basic Local Alignment Search Tool (BLAST) search. Users can set the threshold value for the detection of AMR genes, by default tool searches for 100% identical sequences. The tools fetch the information from other databases like the Marilyn Robers database, Lahey database, and Antibiotic Resistance Database (ARDB; Zankari et al., 2012).

2.5 | Pathway-Systems Resource Investigation Centre (PATRIC)

PATRIC (https://www.patricbrc.org/) database serves as the data repository for different pathogenic data integrated from different types of data types. Various data types include protein–protein interactions (PPI), three-dimensional (3D) protein structures, bacterial genomics, transcriptomics, and sequencing typing data. PATRIC provides the user with an online resource for retrieving the summarized individual genomic data and as well as the gene information. PATRIC also contains the tools for comparative genomics and transcriptomic analysis. It is a joint effort of the National Institute of Allergy and Infectious Disease-funded Bioinformatics Resource Center (BRC). The gene annotation in the PATRIC database are regularly updated using Rapid Annotations using Subsystems Technology (RAST; Wattam et al., 2014, 2017).

2.6 | Antibiotic resistance database (ARDB)

ARDB (http://ardb.cbcb.umd.edu/) database is a data repository for manually curated data of AR genes in pathogenic bacteria. In the ARDB database, the gene entries are listed along with their AR profiles, mechanism of drug resistance, Clusters of Orthologous Groups, and Conserved Domain Database annotations. ARDB also provides useful external links to protein sequence databases (Liu & Pop, 2009).

2.7 | Comprehensive antibiotic resistance database (CARD)

CARD (https://card.mcmaster.ca/) database is one of the dedicated databases for manually curated AR gene data with high-quality reference data on the molecular basis of each AR gene. It also provides the molecular sequence analysis tools such as BLAST and the Resistance Gene Identifier (Jia et al., 2017; McArthur et al., 2013).

2.8 | National database of antibiotic resistance organisms (NDARO)

NDARO (https://www.ncbi.nlm.nih.gov/pathogens/antimicrobial-resistance/) serves as the data repository for curating and presenting the AR gene information in pathogenic bacteria. It is hosted and maintained by the NCBI.

2.9 | Search tool for the retrieval of interacting genes/proteins (STRING)

STRING (https://string-db.org/) database is one among the well-known protein–protein interaction databases along with the Biological General Repository for Interaction Datasets, the biomolecular interaction Network Database, IntAct molecular interaction database, database of interacting proteins, Molecular INTeraction database, and Human Protein Reference Database. The STRING database serves as the data repository for known and as well as the predicted PPI, which includes both direct and indirect PPI. Direct interactions are physical interactions that have the experimental evidence proved by the lab techniques, whereas the indirect interactions are predictions made using computational approaches. The data curated in the STRING includes data from high-throughput experiments, data from data mining approaches, data from the available scientific literature, and also from the gene coexpression studies. STRING provides a confidence score or combined scores for every interaction in the database, based on the available source of data curation. The confidence scores lie between 0 and 1; the more the evidence, the more the confidence scores. String classifies the interactions into four categories based on the confidence scores, highest (above 0.90), high (0.7–0.89), medium (0.4–0.69), and low (0.15–0.39) (Szklarczyk et al., 2011, 2015).

2.10 | The database for annotation, visualization, and integrated discovery (DAVID)

DAVID database (https://david.ncifcrf.gov/) provides the user with a set of tools useful for the functional annotation of the genes. DAVID tools can help the user to identify the gene ontology (GO) terms, functional over-represented gene groups, and also visualize the genes in pathway databases such as BioCarta and Kyoto Encyclopedia of Genes and Genomes (KEGG). It also consists of the external links to various databases associated with the classification of the genes (Huang et al., 2009; Jiao et al., 2012).

2.11 | Cytoscape

Cytoscape is an open-source visualization and network analysis software used to construct and analyze biological and social interaction networks. Cytoscape comes with core features for data integration, visualization, and network analysis. Additional features added for free via Cytoscape plugins available in the Cytoscape app store. The additional plugins include apps related to new layouts, file format conversions, scripting, and molecular profiling analysis. The data from other databases can obtain by installing the integrated database plugins (Demchak et al., 2014; Saito et al., 2012). In biological networks, every entity (protein/gene) called a node and connections between the nodes called edges.

2.11.1 | Molecular complex detection (MCODE)

MCODE (http://baderlab.org/Software/MCODE) plugin is an add-on app to the Cytoscape used for identifying the highly interconnecting nodes (also called as clusters) in an interaction network. The clustered nodes are more or likely to have a functional similarity. The MCODE algorithm mainly depends on scoring the number of interacting nodes (node scoring), identification of possible clusters, and post-processing of dense interacting regions where the nodes with more interacted nodes in a network. MCODE assigns scores for each cluster and ranks based on the density and size of the cluster (Bader & Hogue, 2003).

2.11.2 | NetworkAnalyzer

NetworkAnalyzer (http://med.bioinf.mpi-inf.mpgde/networkanalyzer/) was used for identifying the topological parameters of the nodes in a network. The tool can efficiently compute the simple and complex topological parameters of both directed and undirected interaction networks. Topological parameters include the number of nodes, connecting components of each node, diameter, radius, and density of the network, number of neighbors for each node, clustering coefficient, average shortest path length and various centralities such as betweenness centrality and closeness centrality (Assenov, Ramírez, Schelhorn, Lengauer, & Albrecht, 2008).

2.12 | Therapeutic target database (TTD)

TTD database (http://bidd.nus.edu.sg/group/cjttd/) provides the information of known and explored therapeutic proteins and also the targeted nucleic acids related to disease along with the critical pathways. TTD also provides the links related to the drug target function, sequence, 3D structure along with the enzyme nomenclature, drug structure, and the classification (Y. Wang et al., 2020).

3 | RESULTS AND DISCUSSION

3.1 | Phylogenetic tree construction and analysis

The whole genome sequence of *P. mirabilis* HI4320, a representative strain of the *P. mirabilis* (Letunic & Bork, 2019; Pearson et al., 2008) was retrieved along with the 28 whole-genome sequences of *P. mirabilis* strains from the NCBI-Genome database. Phylogenetic analysis was done using ParSNP and iTOL to understand the evolutionary relationship among the *P. mirabilis* strains. iTOL tool provides the phylogenetic tree either as the rooted tree or as a circular tree (Figure 1). From the phylogenetic tree, we have observed that the strain *P. mirabilis* HI4320 has close relatedness with the strains *P. mirabilis* ENT1224, *P. mirabilis* ENT1157, *P. mirabilis* T18, *P. mirabilis* T21, *P. mirabilis* AOUC001, *P. mirabilis* CYPV1, and *P. mirabilis* CYPM1 strains. Previous reports on a proteomic comparison of reference strain *P. mirabilis* HI4320 with *P. mirabilis* AOUC001 and *P. mirabilis* CYPM1 has confirmed that they are 99% identical (Armbruster, Mobley, & Pearson, 2018). The AMR genes, along with their drug resistance profile in each *P. mirabilis* strain, have been collected from the ResFinder database and are provided in Table S1. We have also obtained the AMR genes reported in various *P. mirabilis* strains from the databases PATRIC, NDARO, CARD, and the AMR gene data was used for gene interaction network construction and analysis.

3.2 | Interaction data curation and network construction

The ResFinder database and the AMR databases have redundant data related to various AR genes. After filtering the redundant entries, a list of 105 unique AMR genes were extracted, and the interaction data related to AMR genes along with their functional partners were retrieved from STRING. Out of 105 genes, the interaction data was available for only 49 genes, so we further extended the network to get the maximum number of interactions between AMR genes and interacting functional partners. The final set of interaction data consists of 74 genes with 793 functional interactions at a medium confidence score. Among the 793 functional interactions, 618 interactions were observed to have the highest confidence scores, 55 interactions were with high confidence scores, and 120 interactions with medium confidence scores. Clustering and network analysis were performed using Cytoscape.

3.3 | Network analysis

Cytoscape-NetworkAnalyzer was used to analyze the topological parameters of the genes in the interaction network. The NetworkAnalyzer results are tabulated and provided in the Table S2. We observed that among the 74 nodes, six nodes have more than 40 direct interactors followed by 31 nodes with 30–40 interactors, six nodes with 10–30 interactors, and 31 nodes with less than 10 direct interactors. The list of top 10 genes with more number of functional interactors is listed in Table 1. In the network, the genes with more direct interactions are considered as the hub molecules, and these genes are considered as the controlling points of molecular interactions. The hub genes can help in better understanding the molecular events and can be exploited as the therapeutic targets for new drug discovery. The top six genes with the highest value are *rpoB*, *PMI2792 (tufB)*, *rpsl*, *fusA*, *rpoC*, and *rpoA*. The genes *rpoA*, *rpoB*, and *rpoC* were involved in the transcription of DNA into RNA. The *tufB* gene plays a crucial role in protein biosynthesis by promoting the

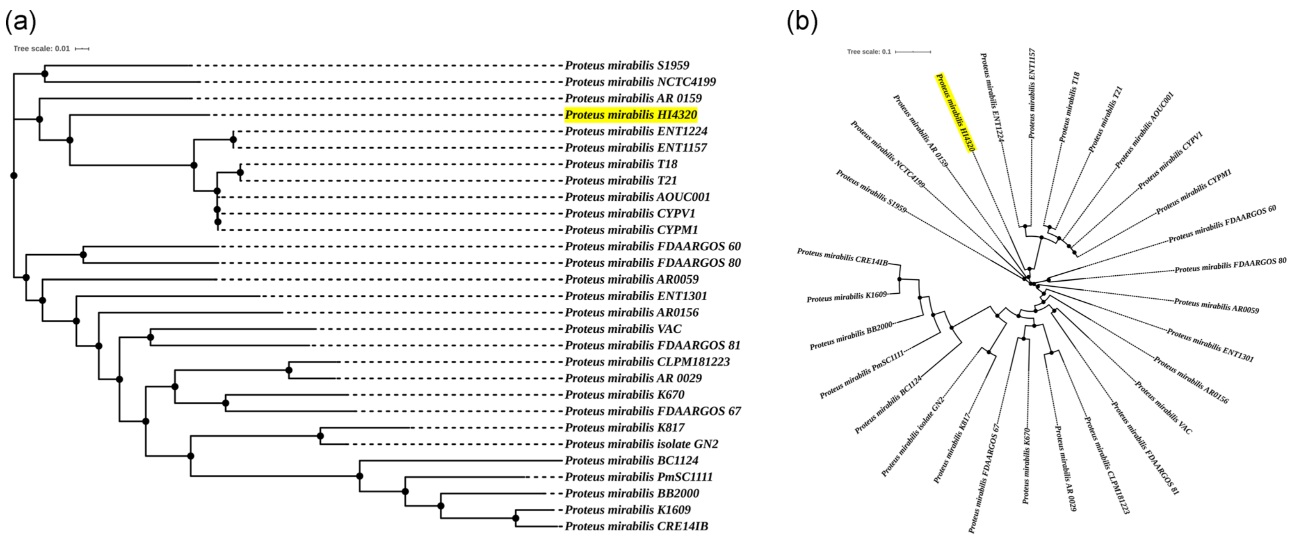


FIGURE 1 Phylogenetic tree analysis of *Proteus mirabilis* strains: 29 whole genome sequences available for *P. mirabilis* strains are downloaded from NCBI-Genome database. The strain *P. mirabilis* HI4320 (highlighted in yellow) is taken as the reference sequence as it is considered as the representative of *P. mirabilis*. The phylogenetic tree was constructed using Parsnp tool and visualized using iTOL tool. (a) Phylogenetic tree in rooted tree view. (b) Phylogenetic tree in circular tree view

TABLE 1 The gene interaction network of AMR genes of *Proteus mirabilis* HI4320 along with the functional partners

Sl. no.	Gene name	Degree	Functional partners
1	*rpoB*	46	*crp, ddlA, folA, folP, fusA, gyrA, gyrB, ileS, macB, murA, parE, PMI2792, rho, rplA, rplB, rplC, rplD, rplE, rplF, rplK, rplM, rplN, rplO, rplP, rplR, rplV, rplW, rplX, rpmD, rpoA, rpoC, rpsB, rpsC, rpsD, rpsE, rpsG, rpsH, rpsI, rpsJ, rpsK, rpsl, rpsM, rpsN, rpsQ, rpsS, secY*
2	*PMI2792*	42	*cpxR, folP, fusA, gyrA, ileS, parE, rplA, rplB, rplC, rplD, rplE, rplF, rplK, rplM, rplN, rplO, rplP, rplR, rplV, rplW, rplX, rpmD, rpoA, rpoB, rpoC, rpsB, rpsC, rpsD, rpsE, rpsG, rpsH, rpsI, rpsJ, rpsK, rpsl, rpsM, rpsN, rpsQ, rpsS, secY, tetAJ, tolC*
3	*rpsl*	41	*arnA, fusA, gidB, gyrA, gyrB, ileS, macB, PMI2792, rplA, rplB, rplC, rplD, rplE, rplF, rplK, rplM, rplN, rplO, rplP, rplR, rplV, rplW, rplX, rpmD, rpoA, rpoB, rpoC, rpsB, rpsC, rpsD, rpsE, rpsG, rpsH, rpsI, rpsJ, rpsK, rpsM, rpsN, rpsQ, rpsS, secY*
4	*fusA*	40	*fabF, gyrA, gyrB, ileS, PMI2792, rho, rplA, rplB, rplC, rplD, rplE, rplF, rplK, rplM, rplN, rplO, rplP, rplR, rplV, rplW, rplX, rpmD, rpoA, rpoB, rpoC, rpsB, rpsC, rpsD, rpsE, rpsG, rpsH, rpsI, rpsJ, rpsK, rpsl, rpsM, rpsN, rpsQ, rpsS, secY*
5	*rpoC*	40	*crp, fusA, gyrA, gyrB, ileS, PMI2792, rho, rplA, rplB, rplC, rplD, rplE, rplF, rplK, rplM, rplN, rplO, rplP, rplR, rplV, rplW, rplX, rpmD, rpoA, rpoB, rpsB, rpsC, rpsD, rpsE, rpsG, rpsH, rpsI, rpsJ, rpsK, rpsl, rpsM, rpsN, rpsQ, rpsS, secY*
6	*rpoA*	40	*crp, fusA, gyrA, gyrB, ileS, PMI2792, rho, rplA, rplB, rplC, rplD, rplE, rplF, rplK, rplM, rplN, rplO, rplP, rplR, rplV, rplW, rplX, rpmD, rpoB, rpoC, rpsB, rpsC, rpsD, rpsE, rpsG, rpsH, rpsI, rpsJ, rpsK, rpsl, rpsM, rpsN, rpsQ, rpsS, secY*
7	*rpsE*	39	*fusA, gyrA, gyrB, ileS, parE, PMI2792, rplA, rplB, rplC, rplD, rplE, rplF, rplK, rplM, rplN, rplO, rplP, rplR, rplV, rplW, rplX, rpmD, rpoA, rpoB, rpoC, rpsB, rpsC, rpsD, rpsG, rpsH, rpsI, rpsJ, rpsK, rpsl, rpsM, rpsN, rpsQ, rpsS, secY*
8	*rpsK*	38	*fusA, gyrA, gyrB, ileS, PMI2792, rplA, rplB, rplC, rplD, rplE, rplF, rplK, rplM, rplN, rplO, rplP, rplR, rplV, rplW, rplX, rpmD, rpoA, rpoB, rpoC, rpsB, rpsC, rpsD, rpsE, rpsG, rpsH, rpsI, rpsJ, rpsl, rpsM, rpsN, rpsQ, rpsS, secY*
9	*rplB*	38	*fusA, gyrA, gyrB, ileS, PMI2792, rplA, rplC, rplD, rplE, rplF, rplK, rplM, rplN, rplO, rplP, rplR, rplV, rplW, rplX, rpmD, rpoA, rpoB, rpoC, rpsB, rpsC, rpsD, rpsE, rpsG, rpsH, rpsI, rpsJ, rpsK, rpsl, rpsM, rpsN, rpsQ, rpsS, secY*
10	*rpsJ*	38	*fusA, gyrA, gyrB, PMI2792, rho, rplA, rplB, rplC, rplD, rplE, rplF, rplK, rplM, rplN, rplO, rplP, rplR, rplV, rplW, rplX, rpmD, rpoA, rpoB, rpoC, rpsB, rpsC, rpsD, rpsE, rpsG, rpsH, rpsI, rpsK, rpsl, rpsM, rpsN, rpsQ, rpsS, secY*

Note: A total of 74 genes with 793 functional interactions were curated from the STRING database. The top 10 genes with more number of direct interactions (degree) along with their respective functional interactions were listed.

guanosine triphosphate (GTP)-mediated binding of aminoacyl transfer RNA (tRNA) to the A-site of ribosomes. *fusA* gene plays an essential role in the GTP-mediated ribosomal translocation of the translation elongation process. *rpsL* interacts and stabilizes the bases of 16S ribosomal RNA (rRNA), which involved in tRNA selection in the A site along with the messenger RNA backbone. Inhibition of *rpoB*, *tufB*, *rpsl*, *fusA*, and *rpoA* genes could bring about the disruption of the network, and thereby the bacterium would find it difficult to synthesize proteins essential for exhibiting AR.

The top 20 genes obtained from the network analysis with the least average shortest path length, and the highest closeness centrality values are listed in Table 2. Important topological parameters, such as betweenness centrality and shortest path length were observed in the data set. Betweenness centrality of a node indicates the importance of the node's centrality in a network. It implies that the number of shortest paths from all other routes to other nodes that pass through this particular node, whereas the shortest path length describes the shortest distance between any two interacting nodes in a network (Bader, Kintali, Madduri, & Mihail, 2007).

We used the TTD database to retrieve the resistance profiles of the top 20 genes with high degree. We observed that out of 20 genes, two genes *rpoB* and *fusA* has shown entries related to two approved drugs, Rifamycin and Fusidic acid. Rifamycin and Fusidic acid are the recommended antibiotics for treatment of most bacterial infections including *P. mirabilis.* Previous reports on antimicrobial susceptibility tests on *P. mirabilis* strains have reported that *P. mirabilis* strains are naturally susceptible to Rifamycin and Fusidic acid (Armbruster et al., 2018). But due to the presence of AMR genes, *P. mirabilis* has acquired the resistance against these antibiotics (Saeb, Al-Rubeaan, Abouelhoda, Selvaraju, & Tayeb, 2017; Sohail, Khurshid, Murtaza Saleem, Javed, & Khan, 2015). We analyzed the drug resistance patterns of each *P. mirabilis* strain included in our study using ResFinder. We noticed that the reference strain *P. mirabilis* HI4320, *P. mirabilis* ENT1224, *P. mirabilis* ENT1157, *P. mirabilis* CYPV1 and *P. mirabilis* CYPM1 strains exhibit resistance to phenicols (*cat* gene) and tetracyclines (*tet(J)* gene). Whereas the strains *P. mirabilis* T18, *P. mirabilis* T21, and *P. mirabilis* AOUC001 exhibit resistance towards aminoglycosides, beta-lactams, phenicols, sulphonamides, tetracyclines, and trimethoprim. *P. mirabilis* AOUC001 has acquired resistance to macrolides and quinolones.

3.4 | Clustering analysis

Clustering analysis of the 74 genes in the network was performed using Cytoscape-MCODE. MCODE revealed four densely interconnected gene clusters based on the number of direct interactions and the connectivity of the genes in the network. Out of 74 genes 50 genes were found in four clusters and are labeled as cluster C1–C4 for easy identification (Table 3). Among the four clusters, the cluster C1 has shown dense interactions with 36 genes and 630 functional interactions (Figure 2). Whereas the clusters C2, C3, and C4 have shown 7, 4, and 3 genes, with 12, 6, and 3 edges, respectively (Figure 3).

TABLE 2 Topological parameter analysis using NetworkAnalyzer: All the genes in the network are analyzed for the topological parameters

Sl. no.	Gene name	Avg. shortest path length	Clustering coefficient	Closeness centrality	Betweenness centrality	Neighbourhood connectivity
1	*rpoB*	1.479452	0.645411	0.675926	0.20198	30.84783
2	*PMI2792*	1.575342	0.75029	0.634783	0.092203	33.16667
3	*rpsI*	1.547945	0.792683	0.646018	0.126517	33.85366
4	*rpoC*	1.726027	0.842308	0.579365	0.009485	34.475
5	*rpoA*	1.726027	0.842308	0.579365	0.009485	34.475
6	*fusA*	1.712329	0.839744	0.584	0.057439	34.45
7	*rpsE*	1.726027	0.877193	0.579365	0.006648	35.33333
8	*rpsK*	1.753425	0.918919	0.570313	0.001879	36
9	*rplB*	1.753425	0.918919	0.570313	0.001879	36
10	*rpsC*	1.753425	0.918919	0.570313	0.001879	36
11	*rpsJ*	1.753425	0.911807	0.570313	0.002678	35.86842
12	*secY*	1.753425	0.911807	0.570313	0.002678	35.86842
13	*rplM*	1.643836	0.901849	0.608333	0.027551	35.97368
14	*rplA*	1.767123	0.948949	0.565891	0.001215	36.59459
15	*rpsI*	1.767123	0.948949	0.565891	0.001215	36.59459
16	*rplD*	1.767123	0.947447	0.565891	0.00105	36.51351
17	*rplR*	1.753425	0.933934	0.570313	0.004871	36.40541
18	*rpsB*	1.767123	0.927928	0.565891	0.027783	36.13514
19	*rplK*	1.780822	0.980952	0.561538	0.000386	37.13889
20	*rplN*	1.780822	0.980952	0.561538	0.000386	37.13889

Note: The top 20 genes with the least average shortest path length and the highest closeness centrality values are listed. The average shortest path length gives the expected distance between the two connected nodes and genes with shortest path length and high closeness centrality are considered as the controlling points of molecular communication. The cluster coefficient value lies in between 0 and 1. The value 0 indicates the node with less than 2 neighbors.

3.5 | Functional enrichment analysis

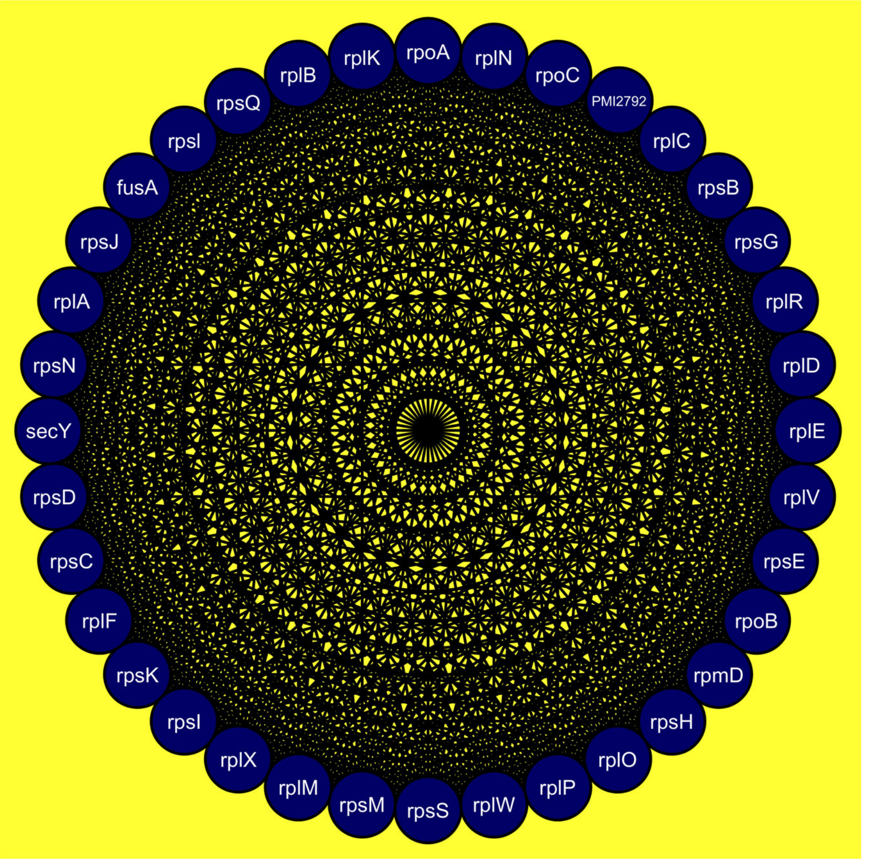
The functional enrichment analysis of the *P. mirabilis* AMR genes along with the functional partners in the network was performed using STRING and the entries are listed in Table S3. Functional enrichment analysis was done to capture the maximum information related to the functional association of these AMR genes at the molecular level. We retrieved 9,598 entries related to the 74 genes (Table S4) from DAVID database. We observed that the GO terms such as biological processes (BP), molecular functions (MF), and

TABLE 3 Clustering analysis of *Proteus mirabilis* HI4320 AMR genes along with their functional partners

Cluster	Score = Density × no. of nodes	Nodes	Edges	Node IDs
C1	36	36	630	*rpsM, rplN, rplP, rplK, rpmD, rplC, rpsD, fusA, rpsN, rpsI, rpsG, rpoC, rplX, rplA, rplM, rplO, rpsB, PMI2792, rpsQ, rpsC, rpoA, rpsH, rpsI, rplB, rplW, rplD, rplE, rpsS, rpsK, rplR, rpsE, rpsJ, rplF, rplV, secY, rpoB*
C2	4	7	12	*mdtC, macB, macA, acrB, emrB, mdtB, mdtA*
C3	4	4	6	*arnA, arnC, arnB, arnT1*
C4	3	3	3	*tetAJ, folP, cpxR*

Note: Cytoscape-MCODE identifies the highly interacting nodes present in the network. Four potential gene clusters are identified and names Cluster C1–C4.

FIGURE 2 Clustering analysis of antimicrobial resistance genes in *Proteus mirabilis* HI4320: Cytoscape-MCODE tool has shown four densely interconnected nodes. The cluster C1 has shown 36 nodes and 630 edges with MCODE score 36



cellular components (CC), KEGG pathways related to various drug resistance mechanisms were enriched. The GO terms, BP's are the processes that are important for the survival of the cell and describe the capacity of various cellular interactions. MF's are the essential activities of the gene product at the molecular level, such as binding or catalysis. CC's are the biomolecular complexes and the structures of the cell (Hill, Smith, McAndrews-Hill, & Blake, 2008).

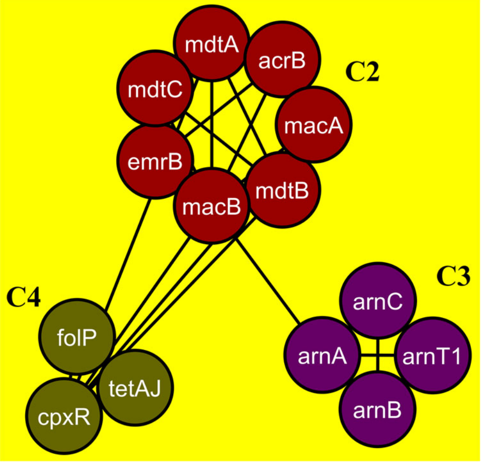


FIGURE 3 Clustering analysis of antimicrobial resistance genes in *Proteus mirabilis* HI4320 using Cytoscape-MCODE: The cluster C2–C4 does not show dense interactions as cluster C1. The cluster C2 has seven nodes and 12 edges followed by cluster C3 with four nodes and six edges. The cluster C4 has three nodes and three edges

The enriched BP entries include the cellular biosynthetic process (GO: 0044249), organic substance biosynthetic process (GO: 1901576), cellular macromolecule biosynthetic process (GO: 0034645), primary metabolic process (GO: 0044238), and cellular protein metabolic process (GO: 004267). In addition to the above metabolic pathways it is noteworthy to mention that pathway related to Vancomycin resistance (GO: 0046677) was also enriched. During the antibiotic exposure, the bacterium reprograms its metabolic network by balancing the biosynthetic processes using the biogenesis of ATP to support bacterial growth and survival in the host. Due to their importance in bacterial growth and survival, the genes related to cellular metabolic pathways are considered as promising drug targets (Murima, McKinney, & Pethe, 2014). In our present study, 37 genes *arnA*, *arnB*, *arnC*, *arnT1*, *ddlA*, *dxr*, *fusA*, *ileS*, *murA*, *rplA*, *rplB*, *rplC*, *rplD*, *rplE*, *rplF*, *rplK*, *rplM*, *rplN*, *rplO*, *rplP*, *rplR*, *rplV*, *rplW*, *rplX*, *rpmD*, *rpoB*, *rpsB*, *rpsD*, *rpsG*, *rpsH*, *rpsI*, *rpsJ*, *rpsK*, *rpsN*, *rpsQ*, *rpsS*, and *rpsI* were observed in the enriched BP related to the cellar metabolic processes (Figure 4). In general, the Gram-negative bacteria intrinsically show resistance to Vancomycin due to the outer membrane impermeability to the large glycol-peptide molecules. We found the genes *alr*, *dadB*, and *ddlA* were enriched with Vancomycin drug resistance (Kerr, 2004). It is interesting to note that the genes

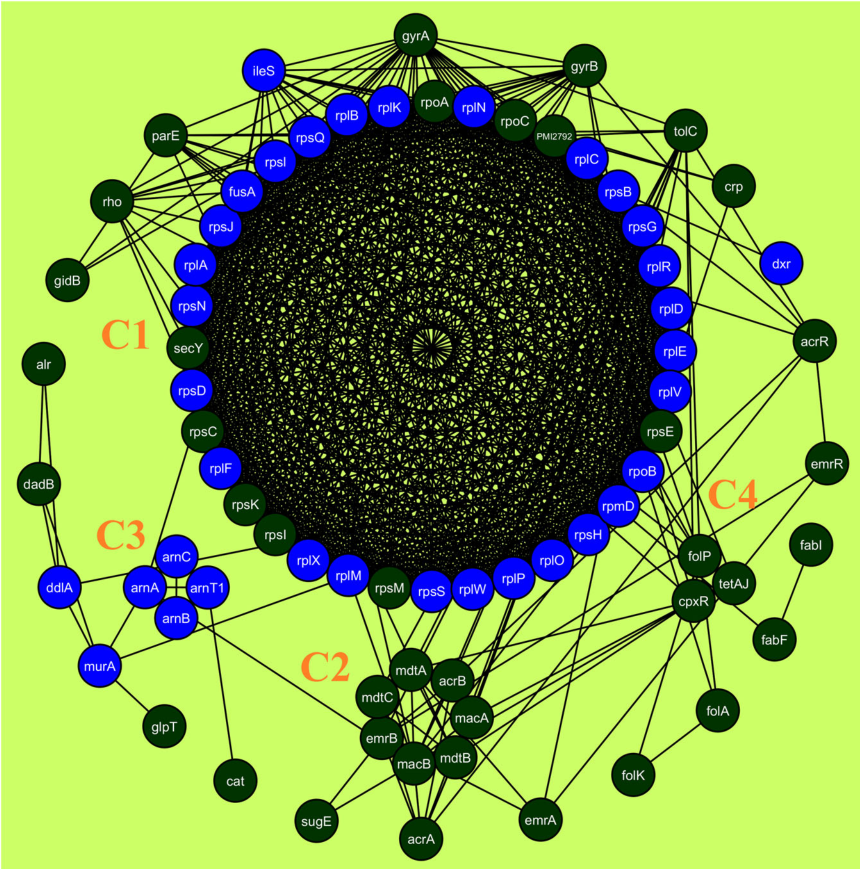


FIGURE 4 The antimicrobial resistance genes of *P. mirabilis* HI4320 enriched in BPs: Functional enrichment analysis was performed using STRING and DAVID tools. The 37 genes were enriched in various BPs were highlighted in Blue color. BP, biological processes

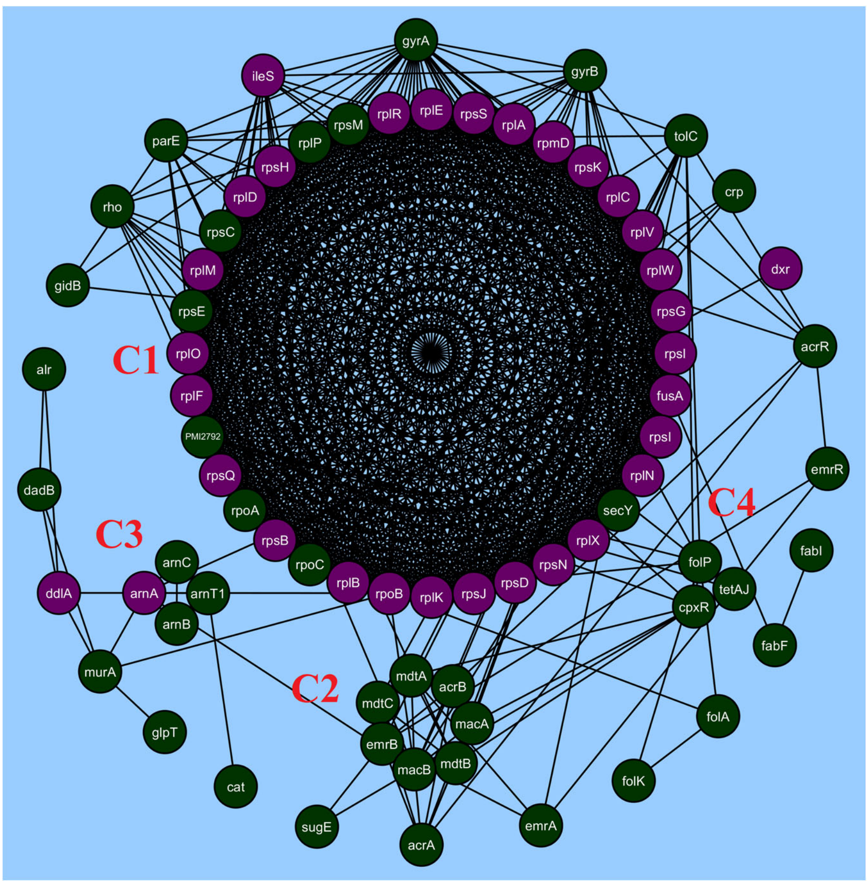


FIGURE 5 The antimicrobial resistance genes of *Proteus mirabilis* HI4320 enriched in MFs: Functional enrichment analysis was performed using STRING and DAVID tools. Out of 74 genes in the network, there are only 33 genes enriched in various MFs. The genes enriched in MFs were highlighted in Violet color. MF, molecular functions

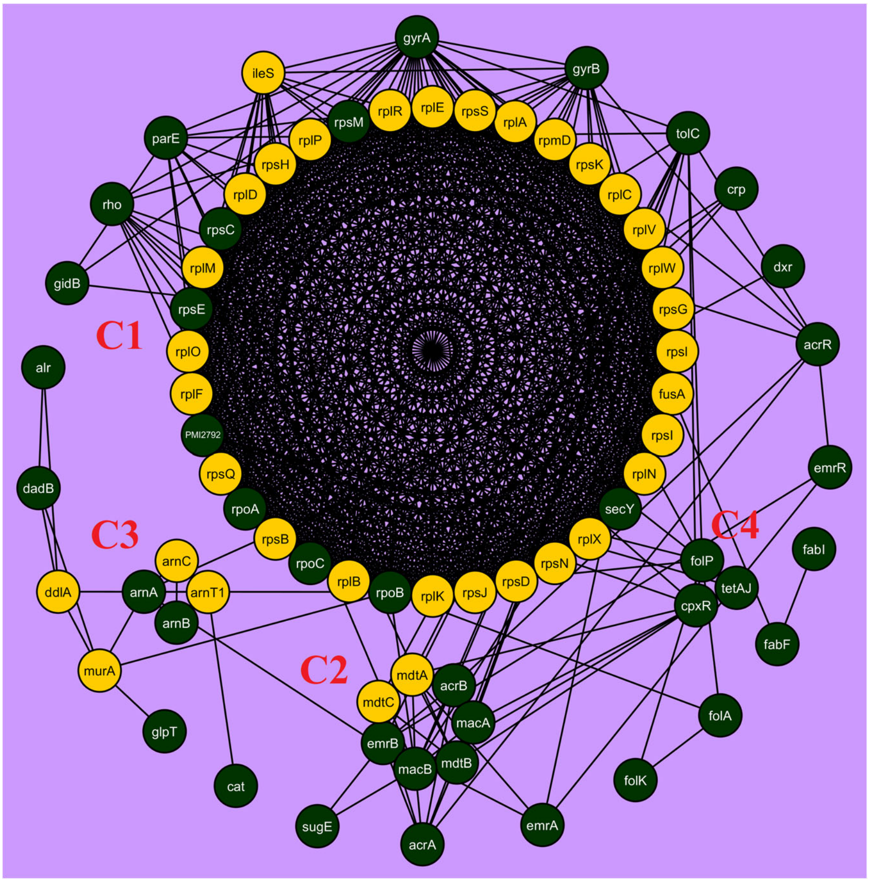


FIGURE 6 The antimicrobial resistance genes of *Proteus mirabilis* HI4320 enriched in CCs: Functional enrichment analysis was performed using STRING and DAVID tools. The genes enriched in various CCs were highlighted in Yellow color. Out of 74 genes there are only 35 genes enriched in CCs. CC, cellular components

arnA, *arnB*, and *arnC* were enriched for the BP, response to antibiotics. These three genes belong to *arnBCADTEF* operon, which was known to have a role in Polymyxin resistance in *P. mirabilis* (Olaitan, Morand, & Rolain, 2014). Other significant result includes the enrichment of MF's such as organic cyclic compound binding (28 genes), structural constituent of ribosome (27 genes), nucleic acid binding (26 genes), RNA binding (25 genes), and rRNA binding (22 genes). A total of 33 genes were enriched in eight MFs enriched in functional enrichment analysis (Figure 5). Out of 74 genes, 35 genes are enriched in six CCs observed in our study (Figure 6). The enriched CC entries include the enrichment of entries such as cell part (35 genes), cytoplasm (31 genes), ribosome (27 genes), and plasma membrane (four genes).

Significant results observed from our study includes the enrichment of KEGG pathways such as ribosome (pmr03010), two-component system (pmr02020), Cationic antimicrobial peptide (CAMP) resistance (pmr015303) and Vancomycin resistance (pmr01502). A total of 52 genes were enriched in nine KEGG pathways (Figure 7). CAMP's are an essential component of host innate immune defense mechanisms against a variety of pathogens, among which the Gram-negative bacteria represent the principal target. CAMP's plays a vital role in the inhibition of bacterial colonization and also in the clearance of infection. They are a large group of peptides produced by various organisms ranging from prokaryotes to vertebrates. Polymixins are one among the CAMPs belonging to the class of antibiotics derived from the Gram-positive bacterium *Bacillus polymyxa* and used for the treatment of multidrug resistant Gram-negative bacteria. Pathogenic Gram-negative bacteria have evolved to avoid killing by host immune system and also by the polymixin antibiotics (Band & Weiss, 2014). The *P. mirabilis* genes *acrA*, *acrB*, *arnA*, *arnB*, *arnC*, *arnT1*, *cpxR*, and *tolC* were observed to have a role in CAMP resistance. Bacterial two-component system (TCS) proteins play a significant role in cell signaling, adaptation to the environment, and also in pathogenesis. Due to its essential role TCS proteins are considered as potential drug targets (S. Wang, 2012). Seven genes of *P. mirabilis* (*arnB*, *cpxR*, *crp*, *mdtA*, *mdtB*, *mdtC*, and *tolC)* were observed to have a role in TCS mechanism. Other important observations from the study include the enrichment of the KEGG pathway related to beta-lactam resistance, Vancomycin resistance, and Folate biosynthesis. The wild-type *P. mirabilis* strains are usually susceptible to Ampicillin and other β-lactam antibiotics; but due to the progressive increase of resistance for β-lactam antibiotics via the production of β-lactamases, *P. mirabilis* has developed resistance to extended-spectrum β-lactam antibiotics (Aragón et al., 2008; Pagani et al., 2002). The genes *acrA*, *acrB*, and *tolC* were enriched in the β-lactamase related pathway. Reports from literature indicate that these genes help in the drug expulsion from the cell using an acrAB-tolC multidrug efflux pump (Ruzin, Keeney, & Bradford, 2005; Visalli, Murphy, Projan, & Bradford, 2003). Folate biosynthesis pathway is one of the targets for the development of new drug molecules

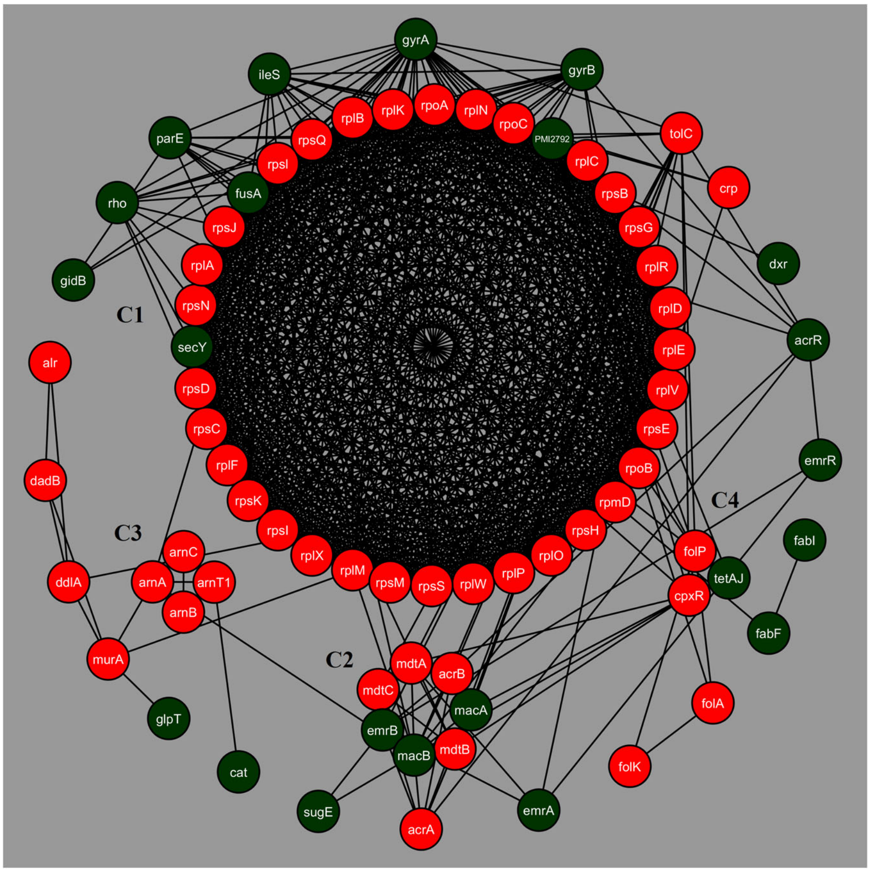


FIGURE 7 The antimicrobial resistance genes of *Proteus mirabilis* HI4320 enriched in KEGG pathways: Functional enrichment analysis was performed using STRING and DAVID tools. The genes enriched in various KEGG pathways were highlighted in Red color. Out of 74 genes there are 52 genes enriched in KEGG pathways. KEGG, Kyoto Encyclopedia of Genes and Genomes

(Bertacine Dias, Santos, Libreros-Zúñiga, Ribeiro, & Chavez-Pacheco, 2018; Bourne, 2014). The genes *folA*, *folk*, and *folP* were observed to be enriched in folate biosynthesis pathway (pmr00790). In addition to the above results, we have also observed various Interpro domain enriched proteins. The enriched entries are related to RND efflux pump fusion proteins (AcrA, EmrA, MacA, and MdtA) and major facilitator superfamily proteins (Bcr, EmrB, GlpT, TetA, and TetJ).

4 | CONCLUSION

The opportunistic bacterial pathogen *P. mirabilis* is the major causal agent within the genus *Proteus* and is mainly associated with UTI infections in the nosocomial environment. In the present study, we utilized the gene interaction network analysis of AMR genes in *P. mirabilis* to understand the drug resistance patterns. We observed that the genes responsible for protein synthesis had interconnections with antibiotic-resistant genes. Functional enrichment analysis reveal various GO terms related to metabolic biosynthesis, cellular biosynthesis processes, AR pathways related to Vancomycin, and other beta-lactam antibiotics. The clustering analysis divulged densely inter-connected gene sets (Cluster C1–C4). We suggest that the genes *rpoB*, *tufB*, *rpsI*, *fusA*, *rpoC*, and *rpoA* can be considered as the hub nodes as they have more number of direct interactions and they could be more critical in analyzing the molecular-level interactions of the AMR genes. On the whole we perceive that the results presented in our report will be a good starting point for researchers to explore new treatment strategies to control *P. mirabilis* infections.

ACKNOWLEDGMENTS

The authors would like to thank Mr. Adhithya Ragunathan for his help during the initial phase of the study. The authors would like to thank the management of VIT for providing the necessary facilities to carry out this study work. The authors gratefully acknowledge the Indian Council of Medical Research (ICMR), Government of India agency for the research grant (IRIS ID: 2019-0810).

CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

AUTHOR CONTRIBUTIONS

S. R. and A. A. designed the study; M. S. K. collected the data and carried out the computation and generated figures; S. R., A. A., and M. S. K. analyzed the data and wrote the manuscript.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

ORCID

Sravan K. Miryala http://orcid.org/0000-0001-7643-5033
Anand Anbarasu http://orcid.org/0000-0003-2216-7488
Sudha Ramaiah http://orcid.org/0000-0002-4800-329X

REFERENCES

Anitha, P., Anbarasu, A., & Ramaiah, S. (2014). Computational gene network study on antibiotic resistance genes of *Acinetobacter baumannii*. *Computers in Biology and Medicine*, *48*(1), 17–27. https://doi.org/10.1016/j.compbiomed.2014.02.009

Anitha, P., Anbarasu, A., & Ramaiah, S. (2016). Gene network analysis reveals the association of important functional partners involved in antibiotic resistance: A report on an important pathogenic bacterium *Staphylococcus aureus*. *Gene*, *575*(2), 253–263. https://doi.org/10.1016/j.gene.2015.08.068

Anitha, P., Bag, S., Anbarasu, A., & Ramaiah, S. (2015). Gene and protein network analysis of AmpC beta lactamase. *Cell Biochemistry and Biophysics*, *71*(3), 1553–1567. https://doi.org/10.1007/s12013-014-0379-5

Aragón, L. M., Mirelis, B., Miró, E., Mata, C., Gómez, L., Rivera, A., … Navarro, F. (2008). Increase in β-lactam-resistant *Proteus mirabilis* strains due to CTX-M- and CMY-type as well as new VEB- and inhibitor-resistant TEM-type β-lactamases. *Journal of Antimicrobial Chemotherapy*, *61*(5), 1029–1032. https://doi.org/10.1093/jac/dkn056

Armbruster, C. E., Mobley, H. L. T., & Pearson, M. M. (2018). Pathogenesis of *Proteus mirabilis* infection. *EcoSal Plus*, *8*(1), 1–123. https://doi.org/10.1128/ecosalplus.ESP-0009-2017

Assenov, Y., Ramírez, F., Schelhorn, S. E., Lengauer, T., & Albrecht, M. (2008). Computing topological parameters of biological networks. *Bioinformatics*, *24*(2), 282–284. https://doi.org/10.1093/bioinformatics/btm554

Bader, D. A., Kintali, S., Madduri, K., & Mihail, M. (2007). Approximating betweenness centrality. *Algorithms and Models for the Web-Graph*, 124–137. https://doi.org/10.1007/978-3-540-77004-6_10

Bader, G. D., & Hogue, C. W. (2003). An automated method for findingmolecular complexes in large protein interaction networks. *Nucleic Acids Research*, *29*(1), 137–140. https://doi.org/10.1093/nar/29.1.137

Bag, S., Ramaiah, S., & Anbarasu, A. (2015a). Fabp4 is central to eight obesity associated genes: A functional gene network-based polymorphic study. *Journal of Theoretical Biology*, *364*, 344–354. https://doi.org/10.1016/j.jtbi.2014.09.034

Bag, S., Ramaiah, S., & Anbarasu, A. (2015b). Network and polymorphic analysis of obesity candidate gene-plin1: A bioinformatics approach. *International Journal of Human Genetics*, *14*(December), 119–129. https://doi.org/10.1080/09723757.2014.11886235

Band, V. I., & Weiss, D. S. (2014). Mechanisms of antimicrobial peptide resistance in Gram-negative bacteria. *Antibiotics*, *4*(1), 18–41. https://doi.org/10.3390/antibiotics4010018

Bertacine Dias, M. V., Santos, J. C., Libreros-Zúñiga, G. A., Ribeiro, J. A., & Chavez-Pacheco, S. M. (2018). Folate biosynthesis pathway: Mechanisms and insights into drug design for infectious diseases. *Future Medicinal Chemistry*, *10*(8), 935–959. https://doi.org/10.4155/fmc-2017-0168

Bourne, C. R. (2014). Utility of the biosynthetic folate pathway for targets in antimicrobial discovery. *Antibiotics*, *3*(1), 1–28. https://doi.org/10.3390/antibiotics3010001

Debroy, R., Miryala, S. K., Naha, A., Anbarasu, A., & Ramaiah, S. (2020). Gene interaction network studies to decipher the multi-drug resistance mechanism in *Salmonella enterica* serovar Typhi CT18 reveal potential drug targets. *Microbial Pathogenesis*, *142*, 104096. https://doi.org/10.1016/j.micpath.2020.104096

Demchak, B., Hull, T., Reich, M., Liefeld, T., Smoot, M., Ideker, T., & Mesirov, J. P. (2014). Cytoscape: The network visualization tool for GenomeSpace workflows. *F1000Research*, *2014*, 1–12. https://doi.org/10.12688/f1000research.4492.2

Guo, S., Aung, K. T., Tay, M. Y. F., Seow, K. L. G., Ng, L. C., & Schlundt, J. (2019). Extended-spectrum β-lactamase-producing *Proteus mirabilis* with multidrug resistance isolated from raw chicken in Singapore: Genotypic and phenotypic analysis. *Journal of Global Antimicrobial Resistance*, *19*, 252–254. https://doi.org/10.1016/j.jgar.2019.10.013

Hammad, M. A., Sulaiman, S. A. S., Aziz, N. A., & Noor, D. A. M. (2019). Prescribing statins among patients with type 2 diabetes: The clinical gap between the guidelines and practice. *Journal of Research in Medical Sciences*, *24*(1), 15. https://doi.org/10.4103/jrms.JRMS

Hammad, M. A., Sulaiman, S. A. S., Aziz, N. A., Noor, D. A. M., Ahmed, S. S., Shariq, A., … Pearson, M. M. (2019). Uropathogens and their antimicrobial resistance patterns: Relationship with urinary tract infections. *International Journal of Health Sciences*, *13*(2), 48–55. https://doi.org/10.7759/cureus.5752

Hill, D. P., Smith, B., McAndrews-Hill, M. S., & Blake, J. A. (2008). Gene Ontology annotations: What they mean and where they come from. *BMC Bioinformatics*, *9*(Suppl. 5), 1–9. https://doi.org/10.1186/1471-2105-9-S5-S2

Huang, D. W., Sherman, B. T., Zheng, X., Yang, J., Imamichi, T., Stephens, R., & Lempicki, R. A. (2009). Extracting biological meaning from large gene lists with DAVID. *Current Protocols in Bioinformatics*, *27*(1), 1–13. https://doi.org/10.1002/0471250953.bi1311s27

Jia, B., Raphenya, A. R., Alcock, B., Waglechner, N., Guo, P., Tsang, K. K., … McArthur, A. G. (2017). CARD 2017: Expansion and model-centric curation of the comprehensive antibiotic resistance database. *Nucleic Acids Research*, *45*(D1), D566–D573. https://doi.org/10.1093/nar/gkw1004

Jiao, X., Sherman, B. T., Huang, D. W., Stephens, R., Baseler, M. W., Lane, H. C., & Lempicki, R. A. (2012). DAVID-WS: A stateful web service to facilitate gene/protein list analysis. *Bioinformatics*, *28*(13), 1805–1806. https://doi.org/10.1093/bioinformatics/bts251

Kerr, J. (2004). Manual of clinical microbiology, 8th edition. *Journal of Clinical Pathology*, *57*(1), 111. https://doi.org/10.1136/jcp.57.1.111-a

Letunic, I., & Bork, P. (2019). Interactive Tree Of Life (iTOL) v4: Recent updates and new developments. *Nucleic Acids Research*, *47*(W1), W256–W259. https://doi.org/10.1093/nar/gkz239

Liu, B., & Pop, M. (2009). ARDB—Antibiotic resistance genes database. *Nucleic Acids Research*, *37*(Suppl. 1), 443–447. https://doi.org/10.1093/nar/gkn656

McArthur, A. G., Waglechner, N., Nizam, F., Yan, A., Azad, M. A., Baylay, A. J., … Wright, G. D. (2013). The comprehensive antibiotic resistance database. *Antimicrobial Agents and Chemotherapy*, *57*(7), 3348–3357. https://doi.org/10.1128/AAC.00419-13

Mihankhah, A., Khoshbakht, R., Raeisi, M., & Raeisi, V. (2017). Prevalence and antibiotic resistance pattern of bacteria isolated from urinary tract infections in Northern Iran. *Journal of Research in Medical Sciences: The Official Journal of Isfahan University of Medical Sciences*, *22*(5), 108. https://doi.org/10.4103/jrms.JRMS_889_16

Miryala, S. K., Anbarasu, A., & Ramaiah, S. (2018). Discerning molecular interactions: A comprehensive review on biomolecular interaction databases and network analysis tools. *Gene*, *642*, 84–94. https://doi.org/10.1016/j.gene.2017.11.028

Miryala, S. K., Anbarasu, A., & Ramaiah, S. (2019). Systems biology studies in Pseudomonas aeruginosa PA01 to understand their role in biofilm formation and multidrug efflux pumps. *Microbial Pathogenesis*, *136* (March), 103668. https://doi.org/10.1016/j.micpath.2019.103668

Miryala, S. K., Anbarasu, A., & Ramaiah, S. (2020a). Evolutionary relationship of penicillin-binding protein 2 coding pena gene and understanding the role in drug-resistance mechanism using gene interaction network analysis. https://doi.org/10.1007/978-981-13-7968-0

Miryala, S. K., Anbarasu, A., & Ramaiah, S. (2020b). Role of SHV-11, a class A β-Lactamase, gene in multidrug resistance among *Klebsiella pneumoniae* strains and understanding its mechanism by gene network analysis. *Microbial Drug Resistance*, https://doi.org/10.1089/mdr.2019.0430

Miryala, S. K., & Ramaiah, S. (2019). Exploring the multi-drug resistance in *Escherichia coli* O157:H7 by gene interaction network: A systems biology approach. *Genomics*, *111*(4), 958–965. https://doi.org/10.1016/j.ygeno.2018.06.002

Mobley, H. L. T., & Chippendale, G. R. (1990). Hemagglutinin, urease, and hemolysin production by Proteus mirabilis from clinical sources. *Journal of Infectious Diseases*, *161*(3), 525–530. https://doi.org/10.1093/infdis/161.3.525

Murima, P., McKinney, J. D., & Pethe, K. (2014). Targeting bacterial central metabolism for drug development. *Chemistry and Biology*, *21*(11), 1423–1432. https://doi.org/10.1016/j.chembiol.2014.08.020

Naha, A., Kumar Miryala, S., Debroy, R., Ramaiah, S., & Anbarasu, A. (2020). Elucidating the multi-drug resistance mechanism of *Enterococcus faecalis* V583: A gene interaction network analysis. *Gene*, *748*, 144704. https://doi.org/10.1016/J.GENE.2020.144704

Nathan, C., & Cars, O. (2014). Antibiotic resistance—Problems, progress, and prospects. *New England Journal of Medicine*, *371*(19), 1761–1763. https://doi.org/10.1056/NEJMp1408040

Olaitan, A. O., Morand, S., & Rolain, J. M. (2014). Mechanisms of polymyxin resistance: Acquired and intrinsic resistance in bacteria. *Frontiers in Microbiology*, *5*, 1–18. https://doi.org/10.3389/fmicb.2014.00643

Pagani, L., Migliavacca, R., Pallecchi, L., Matti, C., Giacobone, E., Amicosante, G., … Rossolini, G. M. (2002). Emerging extended-spectrum β-lactamases in *Proteus mirabilis*. *Society*, *40*(4), 1549–1552. https://doi.org/10.1128/JCM.40.4.1549

Parimelzaghan, A., Anbarasu, A., & Ramaiah, S. (2016). Gene network analysis of metallo beta lactamase family proteins indicates the role of gene partners in antibiotic resistance and reveals important drug targets. *Journal of Cellular Biochemistry*, *117*(6), 1330–1339. https://doi.org/10.1002/jcb.25422

Pearson, M. M., Sebaihia, M., Churcher, C., Quail, M. A., Seshasayee, A. S., Luscombe, N. M., … Mobley, H. L. T. (2008). Complete genome sequence of uropathogenic *Proteus mirabilis*, a master of both adherence and motility. *Journal of Bacteriology*, *190*(11), 4027–4037. https://doi.org/10.1128/JB.01981-07

Ruzin, A., Keeney, D., & Bradford, P. A. (2005). AcrAB efflux pump plays a role in decreased susceptibility to tigecycline in *Morganella morganii*. *Antimicrobial Agents and Chemotherapy*, *49*(2), 791–793. https://doi.org/10.1128/AAC.49.2.791-793.2005

Saeb, A. T. M., Al-Rubeaan, K. A., Abouelhoda, M., Selvaraju, M., & Tayeb, H. T. (2017). Genome sequencing and analysis of the first spontaneous Nanosilver resistant bacterium *Proteus mirabilis* strain SCDR1. *Antimicrobial Resistance and Infection Control*, *6*(1), 1–27. https://doi.org/10.1186/s13756-017-0277-x

Saito, R., Smoot, M. E., Keiichiro Ono, J., Ruscheinski, O., Wang, P.-L., Lotia, S., … Ideker, T. (2012). A travel guide to cytoscape plugins. *Natural Methods*, *9*(11), 1069–1076. https://doi.org/10.1038/jid.2014.371

Schaffer, J. N., & Pearson, M. M. (2015). *Proteus mirabilis* and urinary tract infections. *Microbiology Spectrum*, *3*(5), 139–148. https://doi.org/10.1128/microbiolspec.UTI-0017-2013

Sharma, D., Preston, S. E., & Hage, R. (2019). Emerging antibiotic resistance to bacterial isolates from human urinary tract infections in grenada. *Cureus*, *11*(9), 5752. https://doi.org/10.7759/cureus.5752

Sohail, M., Khurshid, M., Murtaza Saleem, H. G., Javed, H., & Khan, A. A. (2015). Characteristics and antibiotic resistance of urinary tract pathogens isolated from Punjab, Pakistan. *Jundishapur Journal of Microbiology*, *8*(7), https://doi.org/10.5812/jjm.19272v2

Szklarczyk, D., Franceschini, A., Kuhn, M., Simonovic, M., Roth, A., Minguez, P., … Von Mering, C. (2011). The STRING database in 2011: Functional interaction networks of proteins, globally integrated and scored. *Nucleic Acids Research*, *39*(Suppl. 1), 561–568. https://doi.org/10.1093/nar/gkq973

Szklarczyk, D., Franceschini, A., Wyder, S., Forslund, K., Heller, D., Huerta-Cepas, J., … Von Mering, C. (2015). STRING v10: Protein-protein interaction networks, integrated over the tree of life. *Nucleic Acids Research*, *43*(D1), D447–D452. https://doi.org/10.1093/nar/gku1003

Treangen, T. J., Ondov, B. D., Koren, S., & Phillippy, A. M. (2014a). *Rapid Core-Genome alignment and visualization for thousands of intraspecific microbial genomes keywords*. 1–35.

Treangen, T. J., Ondov, B. D., Koren, S., & Phillippy, A. M. (2014b). The harvest suite for rapid core-genome alignment and visualization of thousands of intraspecific microbial genomes. *Genome Biology*, *15*(11), 1–15. https://doi.org/10.1186/s13059-014-0524-x

Visalli, M. A., Murphy, E., Projan, S. J., & Bradford, P. A. (2003). AcrAB multidrug efflux pump is associated with reduced levels of susceptibility to tigecycline (GAR-936) in *Proteus mirabilis*. *Antimicrobial Agents and Chemotherapy*, *47*(2), 665–669. https://doi.org/10.1128/AAC.47.2.665-669.2003

Wang, S. (2012). Bacterial two-component systems: Structures and signaling mechanisms. In C. Huang (Ed.), *Protein phosphorylation in human health* (pp. 439–466). Rijeka: InTech Press. https://doi.org/10.5772/48277

Wang, Y., Zhang, S., Li, F., Zhou, Y., Zhang, Y., Wang, Z., … Zhu, F. (2020). Therapeutic target database 2020: Enriched resource for facilitating research and early development of targeted therapeutics. *Nucleic Acids Research*, *48*(D1), D1031–D1041. https://doi.org/10.1093/nar/gkz981

Wattam, A. R., Abraham, D., Dalay, O., Disz, T. L., Driscoll, T., Gabbard, J. L., … Sobral, B. W. (2014). PATRIC, the bacterial bioinformatics database and analysis resource. *Nucleic Acids Research*, *42*(D1), 581–591. https://doi.org/10.1093/nar/gkt1099

Wattam, A. R., Davis, J. J., Assaf, R., Brettin, T., Bun, C., Conrad, N., … Stevens, R. L. (2017). Improvements to PATRIC, the all-bacterial bioinformatics database and analysis resource. *Nucleic Acids Research*, *45*, 535–542. https://doi.org/10.1093/nar/gkw1017

Zankari, E., Hasman, H., Cosentino, S., Vestergaard, M., Rasmussen, S., Lund, O., … Larsen, M. V. (2012). Identification of acquired antimicrobial resistance genes. *Journal of Antimicrobial Chemotherapy*, *67*(11), 2640–2644. https://doi.org/10.1093/jac/dks261

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

How to cite this article: Miryala SK, Anbarasu A, Ramaiah S. Gene interaction network approach to elucidate the multidrug resistance mechanisms in the pathogenic bacterial strain *Proteus mirabilis*. *J Cell Physiol*. 2020;1–12. https://doi.org/10.1002/jcp.29874